

Learnable parameter = γ, β

non " " = H, σ

Date: _____

Day: _____

Advantages:-

- Stable, faster, Regularizer.
- weight init impact reduce.

Optimizers:-

how to speed training.



$$W_n = W_0 - \eta \frac{\partial L}{\partial W}$$



G.D:- Update rule.

- Batch optimizer or gd.
- Stochastic " " "
- Mini-batch " " "

Challenges:- 1) Learning rate.

G.D fails it.

2) Learning rate scheduling.

3) Local minima.

4) Saddle point.



5) ...

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Next:-

- 1) Momentum
- 2) Adagrad
- 3) NAG
- 4) RMSProp
- 5) Adam

Exponentially weighted moving average:-

E.W is a technique using which
u find trends in a time series
based data.

- time series forecasting.
- financial " "
- Signal processing.
- Deep Learning, while building optimizers.
- Simply u give more focus to
latest data.
- Over-time avg for any point b/m low.

$$V_t = \beta V_{t-1} + (1 - \beta) \theta_t$$

$$\beta \rightarrow 0 < \beta < 1$$

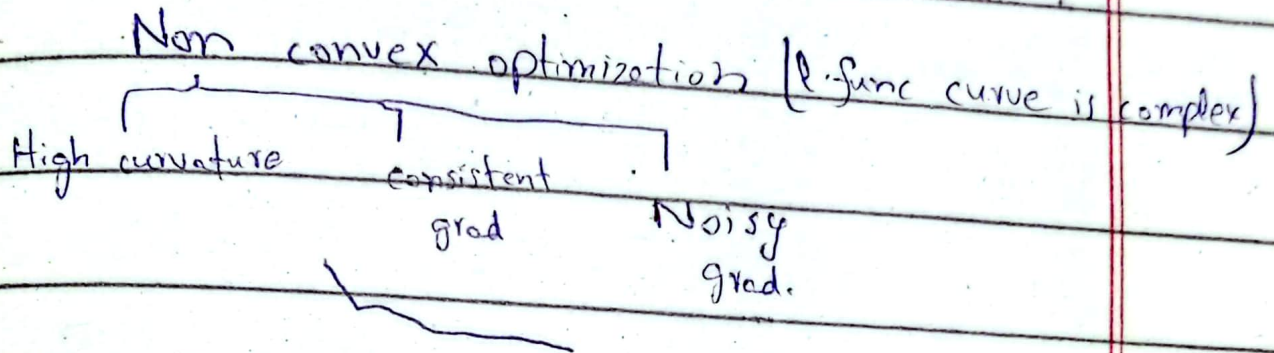
Momentum $\rightarrow$ Speed.



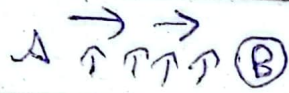
Date: _____

Day: SGD with momentum.

Momentum optimization:-



What:-



if prev grad shifts α in $\perp$ direction, α will go confidently.

$$w_{t+1} = w_t - \eta \text{grad } w_t.$$

history of
past velocity $\rightarrow$ momentum

$v \rightarrow$ velocity.

$$w_{t+1} = w_t - v_t$$
$$v_t = [\beta * v_{t-1}] + \eta \nabla w_t$$
$$0 < \beta < 1$$

Advantages:-

- Speed

escape from local minima.

Disadvantages:-

Oscillation before reaching.

Optimizers are when weights updated loss value minimum optimum.

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Nesterov accelerated gradient:- (NAG):-

Advance form of momentum where oscillations reduced.

Adagrad:- (use when data is sparse).

Learning rate not fixed.

→ changed or adapt accordingly.

→ input features → scale is much different

→ features → sparse → 0 (most)

→ $\ln \uparrow$, grad descent $\downarrow$ & vice versa

Disadvantage: Small updates & can't reach global minima for any epochs.

RMSProp:-

Root mean square propagation
↳ improvement over adagrad.

if there is a convolution layer even if
its just 1 is $\rightarrow$ CNN.

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Adam :-

Adaptive moment estimation.

- $\rightarrow$ idea of momentum
 - $\rightarrow$ learning rate decay.
- } merge

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) \nabla W_t \rightarrow \text{momentum}$$

$$V_t = \beta_2 V_{t-1} + (1 - \beta_2) [\nabla W_t]^2 \rightarrow \text{Adapted}$$